Fiscal Year:	FY 2017	**Task Last Updated:**	FY 07/01/2017
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Project Title:	Amyloid Fibril Formation in Microgravity: Distinguishing Interfacial and Flow Effects		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	BIOPHYSICS--Biophysics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
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Space Biology Special Category:	None		
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Zip Code:	12180-3522	**Congressional District:**	20
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2013 Complex Fluids & Macromolecular Biophysics NNH13ZTT001N
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No. of PhD Candidates:	3	**No. of Master' Degrees:**	2
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	12	**Monitoring Center:**	NASA MSFC
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Formation of amyloid fibrils is widely studied because of their role in neurodegenerative diseases, including Alzheimer's. The fibrillization process is characterized by a nucleation step followed by fibril growth, much like the steps that occur during protein crystallization. Aside from the general interest in macromolecular biophysics, recent evidence shows the relevance of amyloid fibril formation in animal models exposed to cosmic radiation, with obvious implications for interplanetary travel. A preponderance of evidence shows that agitating protein solutions can significantly accelerate amyloid fibril formation; however, the mechanisms by which the presence of bubbles and flow independently affect fibrillization are not known. The situation is further complicated because we do not yet have mechanistic models that can quantify the influence of natural convection due to gravity on the fibrillization process. We know that fluid interfaces (e.g., air-water interface) induce amyloid fibrils when they are sheared, but the influence of

Task Description:	container walls is not fully understood. Here, we aim to distinguish the influence of flow and fluid interfaces on amyloid formation by removing other effects, including solid walls and buoyancy effects. Formation of protein amyloid fibrils provides a model system for studying biomolecular transport phenomena, where the influence of flow can be altogether negated by the elimination of gravity. Furthermore, the experiments can be performed aboard the International Space Station (ISS) with essentially no solid container, since surface tension can provide liquid containment. Thus, the free surface can be readily observed (e.g., with high magnification objectives), manipulated (e.g., can be sheared or left otherwise undisturbed), and modified (e.g., by spreading a monomolecular oil film). First, we will perform experiments at 1g on a small droplet to ascertain the effect of forced convection on a liquid surface. Next, we will generalize our very successful computational model in annular systems to the spherical geometry of liquids at microgravity. We will design the space-based experiments based on the results from the ground experiments and numerical predictions. Finally, we will perform experiments aboard ISS on large-scale drops in an experiment referred to as the Ring-Sheared Drop (RSD).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	When proteins are perturbed from their native state, they lose their native confirmation (folding) and then can bond with other proteins, forming an aggregate. This is the reason why the transparent fluid in an egg becomes a white solid when cooked. Amyloid fibrils are a unique type of protein aggregate that occur naturally in the body. We study them because amyloid plaques in the brain are associated with many neurodegenerative disorders, including Alzheimer's. Fibrils are very intriguing because they are highly ordered and self-assemble. When heated, insulin molecules, for example, denature and bond with other denatured molecules (or assists in denaturing the other molecules). These form a nucleation site, and from there the fibril elongates as more molecule bond into what is called a "beta sheet" structure. The primary mechanism for activation of protein in the body is not yet understood, but it is likely to be a combination of flow friction, aka "shear stress" and interfacial effects that native proteins are exposed to. Aside from their central role in neurodegenerative diseases such as Parkinson's and Alzheimer's, formation of amyloid structures in proteins plays a role in other diseases including diabetes.
Task Progress:	I) Progress in Administration A) The SRD (Science Requirements Document), which was initiated in Year-1 and revised in Years 2 and 3, was further refined in Year-4. Many of the changes to the requirements reflect practical limitations and other considerations revealed during conferences with other scientists and engineers at Marshall Space Flight Center (MSFC) and their prime contractor for the RSD (Teledyne-Brown Engineering). A few other changes to the SRD as a result of the Requirement Document Review (RDR), described as Item-B. The SRD is now ready for signatures. B) The RDR was held in Huntsville AL on February 22, 2017, and successfully passed. In the morning of the RDR, the Principal Investigator (PI)reviewed the scientific motivation behind the RSD project and then detailed the requirements tables in the SRD. The PI and the co-PI also participated in the discussions of the engineering review, presented in the afternoon by Teledyne-Brown Engineering personnel. C) A second parabolic flight opportunity grant application entitled "Protein-Drop Pinning in Microgravity" was submitted in response to NNH16ZOA001N-16FO_F1B. The proposal was successful and an award was made (NNX17AG04G). It provides support for one semester for a graduate student plus cost of hardware, as well as the cost of the flight itself (through the provider: Zero-G Corp.). A contract with Zero-G has been drawn and a purchase order is being processed at Rensselaer Polytechnic Institute (RPI). The contract provides for two people to fly on two separate days as part of a multiday flight campaign scheduled for Nov. 13th through 17th (TTR on the first day). The entire hardware package is being designed and manufactured at RPI. Also, arrangements have been made to provide an opportunity for Teledyne-Brown Engineering team to test their contact ring designs. FO-2 is intended to test deployment and pinning of a 15-mm diameter drop, which is up from the 10-mm drop successfully demonstrated in FO-1 flight campaign in February 2016. Furthermore, FO-2 is exclusively dedicated to study of protein drops, whereas FO-1 (2016) was primarily focused on water drops. To increase chances of success, we were able to secure a commitment from the experienced flyer from FO-1 campaign, Ms. Carole Fritsche (NASA-Kennedy Space Center (KSC) and Stinger Ghaffarian Technologies), to join the RPI team. Two of the students from the FO-1 team, along with two backups, constitute the balance of the FO-2 team. II) Scientific and Technological Progress 1) Extensive studies of flow-induced fibrillization of insulin at room temperature were conducted in a deep-channel surface viscometer and the results were published recently in the journal Soft Matter (Balaraj et al. 2017. Note, this paper was featured as the outside back cover of the March issue). The deep-channel surface viscometer geometry consists of an annular region between stationary cylinders, which is driven by constant rotation of the floor. It was shown that the initial start-up transient in that scale flow (outer radius 2.5 cm) is about a minute. The air–water interface has fibrils on it by about 30 hours after starting the flow (floor angular velocity ~90 RPM), but no sign of fibrillization is detected in the bulk until about two days later. During this time the amount of native protein in solution has only reduced by about 10% from its initial concentration. The interface is fully covered by fibrils by about day two, and then the fibril layer thickens with time. Until fibrils appear on the interface, the interfacial flow is slowly diminishing and the surface shear viscosity slowly increases. Then at about 30 hours after the start-up, coinciding with the appearance of the fibrils on the interface, the interface is completely immobilized and the surface shear viscosity has increased by two orders of magnitude. The rapid growth of the fibril weave observed at the surface was found to be associated with the momentum boundary layer just beneath it. For comparison, the volume-to-surface area ratio for this system was about two times that of the RSD. 2) A predictive model was developed and successfully verified by experiments for non-Newtonian films flowing on Newtonian fluids. This is comparable to the flow protein solution with free surface covered by protein structures such as amyloid fibrils. The model includes two-way (nonlinear) coupling to the bulk flow. The results were published in the Journal of Fluid Mechanics (Underhill et al., 2017). 3) The results of the first campaign of parabolic flights "FO-1" (Feb. 2016, aboard ZeroG Corp. Boeing 727) were published in the journal Microgravity Science and Technology (Gulati et al., 2017), along with laboratory experiments and simulations. The results showed that in microgravity, a 1 cm diameter drop of water can be readily grown in a short time (6 s) and pinned between two rings. 4) Through experiments and numerical simulations, we have demonstrated that oscillatory forcing of a knife edge results

	in a strong streaming flow (in the bulk). Also, the sensitivity of the knife edge viscometer was shown to be enhanced with increasing amplitude and frequency of forcing. The results were published in the journal Physical Review Fluids (Rasheed et al., 2017). 5) Recent experiments have revealed a very distinct and (axially) symmetric pattern of amyloid fibrils formed on glass walls as a result of flow (a notion we have termed “flow directed assembly,” FDA). One of the remarkable features of these assemblies is that the fibrils, which are individually on the order of hundreds of microns in length, are themselves aligned with the direction of wall shear stress, whereas, collectively, they are patterned orthogonal to the wall shear stress. We have successfully patterned protein fibrils up to the centime scale, a capability not yet found in the literature. FDA was observed in both deep-channel surface viscometer geometry as well as a rotating Couette flow with a rotating outer cylinder and floor. Additional experiments and simulations are underway and we plan to submit a manuscript in the near future. 6) Significant technological progress has been made on the design of the RSD as well as a the development and testing of probe and imaging system to study the RDS and its Earth-based analog, namely the knife-edge surface viscometer. This year we have managed to make functional the remote-controlled Leica RXA microscope (a duplicate of the LMM (Light Microscopy Module), on loan from Glenn Research Center (GRC)). We are now in the process of making the Yokogawa confocal head also functional. The goal is to enable real-time confocal imaging of protein amyloid fibrillization at a sheared air-water interface.
Bibliography Type:	Description: (Last Updated: 04/15/2026)
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Journal/Magazine covers	Balaraj VS, Zeng PC, Sanford SP, McBride SA, Raghunandan A, Lopez JM, Hirsa AH. "Cover for 'Surface shear viscosity as a macroscopic probe of amyloid fibril formation at a fluid interface.' " Soft Matter. 2017 Mar 1;13(9):1780-7. PubMed PMID: 28177017 ; back cover http://pubs.rsc.org/en/content/articlelanding/2017/sm/c7sm90038d#!divAbstract , Mar-2017